

GLM 5.2 and the Latest-Generation GLM Models: Inference Sources, Pricing, and Free Access for mindX Integration

TL;DR

- **GLM-5.2 is real and released.** Zhipu AI / Z.ai shipped GLM-5.2 to GLM Coding Plan subscribers on June 13, 2026, opened the standalone pay-per-token API on June 16, and published MIT-licensed open weights on June 16-17. It is a 753B-total / ~40B-active MoE with a 1M-token context window. So "GLM 5.2" exists — you do not need to fall back to GLM-4.6.
- **For mindX, lead with an OpenAI-compatible hosted endpoint.** The cheapest predictable route for a JWT-gated, multi-route API is Z.ai's first-party OpenAI-compatible endpoint (<https://api.z.ai/api/openai/v1>) or an aggregator like OpenRouter/DeepInfra. GLM-5.2 list price is \$1.40 in / \$4.40 out per 1M tokens (cached input \$0.26); cheaper FP4 routes (DeepInfra, GMI, Wafer) blend to ~\$0.72-\$0.95. (TogetherAI +2)
- **Free/cheap options exist but with caveats.** There is no **(: free)** GLM-5.2 slug on OpenRouter, but GLM-4.5-Air ([z-ai/glm-4.5-air:free](https://openrouter.ai/providers/zai-org/models/glm-4.5-air)), GLM-4.7-Flash and GLM-4.5-Flash are genuinely free; new BigModel accounts get **20,000,000 free tokens** ("New users get 20.00 million tokens for free!"); and the MIT weights let you self-host at zero per-token cost (but GLM-5.2 is datacenter-class — ~8×H100/H200). The GLM Coding Plan (from ~\$3-18/mo) is the cheapest high-volume route but is restricted to coding tools, not general API backends.

Key Findings

1. Model version availability (confirmed vs. speculative)

- **GLM-5.2 — CONFIRMED RELEASED.** Released to GLM Coding Plan subscribers June 13, 2026; standalone API June 16, 2026 (per Lushbinary, June 25 2026: "Z.ai turned on the standalone pay-per-token API on June 16, 2026, priced at \$1.40 per million input tokens and \$4.40 per million output tokens"); open weights on Hugging Face ([zai-org/GLM-5.2](https://huggingface.co/zai-org/GLM-5.2)) and ModelScope June 16-17, 2026. Official Z.ai blog "GLM-5.2: Built for Long-Horizon Tasks" published June 17, 2026. (huggingface)
- **Specs (primary-source, official Hugging Face model card + Z.ai blog):**
 - **753B total parameters / ~40B activated (MoE).** The official zai-org card states "Model size: 753B params"; the 40B-active figure is confirmed by the first-party NVIDIA-quantized card ([nvidia/GLM-5.2-NVFP4](https://nvidia.com/en-us/ai-dl/ai-models/glm-5.2-nvfp4/)), which states "753B in total and 40B activated." Note a documented conflict: an arXiv generative-AI survey describes the GLM-5 backbone as "a mixture-of-experts architecture with 744 billion total parameters and 40 billion active parameters per forward pass" — the 744B figure

reflects the GLM-5 foundation, while the official GLM-5.2 card says 753B. Use **753B total / 40B active** as the authoritative GLM-5.2 figure.

- Architecture class `GlmMoeDsaForCausalLM` (tag `glm_moe_dsa`), using DeepSeek Sparse Attention (DSA) plus a new **IndexShare** mechanism that reuses one indexer across every four sparse-attention layers, cutting per-token FLOPs by 2.9× at 1M context. An improved Multi-Token Prediction (MTP) layer raises speculative-decoding acceptance length by ~20% (4.56 → 5.47). Technical report: arXiv 2602.15763 ("GLM-5: from Vibe Coding to Agentic Engineering"); IndexShare paper arXiv 2603.12201. `huggingface`
- **Context window: 1M tokens** ("solid 1M-token context," up from 200K in GLM-5.1). **Max output 128K tokens per response** (per Requesty's GLM-5.2 model page: "glm-5.2 has a context window of 1M tokens, with a maximum output of 128K tokens per response"). Together AI's reseller page lists "256K context / 131,072 output" — this is a reseller error; the official figure is 1M.
- Two thinking-effort levels: **High** and **Max** (OpenRouter exposes these as `high` / `xhigh`, where xhigh maps to max). Thinking is enabled by default. `OpenRouter`
- License: **MIT** ("Pure Open: An MIT open-source license — no regional limits"). Covers the open weights; code repo (github.com/zai-org/GLM-5) is also MIT.
- Supported inference frameworks: transformers, vLLM (0.23.0+), SGLang (0.5.13+), KTransformers, xLLM, Unsloth, llama.cpp (GGUF community builds), plus Ascend NPU paths.
- Text-only (no vision). Benchmarks (self-reported / early third-party): 81.0 on Terminal-Bench 2.1, 62.1% on SWE-bench Pro (vs GPT-5.5's 58.6%, Gemini 3.1 Pro's 54.2%); positioned as the strongest open-source model on long-horizon coding benchmarks (FrontierSWE, SWE-Marathon). Treat benchmark superiority claims as vendor-reported pending independent confirmation. `LLM Stats`
- **The broader GLM lineage (all confirmed, MIT-licensed open weights):** GLM-4.5 / GLM-4.5-Air (Jul 2025), GLM-4.5V (Aug 2025), GLM-4.6 (Sep 30, 2025), GLM-4.7 (Dec 2025), GLM-5 (Feb 11, 2026), GLM-5.1 (Apr 7, 2026), GLM-5.2 (Jun 2026). GLM-4.6V is the current vision model; **GLM-4.6 (355B total params per the zai-org/GLM-4.6 Hugging Face model card, ~32B active, 200K context)** remains a very cost-effective workhorse.

2. Sources of inference and pricing

First-party (Z.ai / BigModel): OpenAI-compatible endpoint `https://api.z.ai/api/openai/v1` and Anthropic-compatible endpoint `https://api.z.ai/api/anthropic`. Mainland: `https://open.bigmodel.cn/api/paas/v4`. Official per-1M-token list prices (USD), from docs.z.ai/guides/overview/pricing: `Avenchat`

Model	Input	Cached input	Output	Context
GLM-5.2	\$1.40	\$0.26	\$4.40	1M
GLM-5.1	\$1.40	\$0.26	\$4.40	~200K
GLM-5	\$1.00	\$0.20	\$3.20	200K
GLM-5-Turbo	\$1.20	\$0.24	\$4.00	—
GLM-4.7	\$0.60	\$0.11	\$2.20	~200K
GLM-4.7-FlashX	\$0.07	\$0.01	\$0.40	~200K
GLM-4.6	\$0.60	\$0.11	\$2.20	200K
GLM-4.5-Air	\$0.20	\$0.03	\$1.10	128K
GLM-4.5-X	\$2.20	\$0.45	\$8.90	—
GLM-4.7-Flash	Free	Free	Free	~200K
GLM-4.5-Flash	Free	Free	Free	128K
GLM-4.6V-Flash	Free	Free	Free	131K (vision)

Web Search built-in tool: \$0.01/use. Cached-input storage is currently "limited-time free." 

Third-party providers (all OpenAI-compatible) — GLM-5.2 and GLM-4.6:

Provider	GLM-5.2 input/output (per 1M)	GLM-4.6 input/output	Context	OpenAI- compat	Notes
Z.ai (first-party)	\$1.40 / \$4.40	\$0.60 / \$2.20	1M / 200K	Yes (+ Anthropic)	Cached input \$0.26; FP8
OpenRouter (router)	list \$1.40 / \$4.40; auto-route ~\$0.95 / \$3.00	~\$0.43 / \$1.74	1M	Yes	22+ providers; no :free for 5.2
DeepInfra	\$0.95 / \$3.00 (FP4)	~\$0.55-0.60 / \$2.20	1M	Yes	Cheapest output; FP4; explicit cached p
GMI Cloud	~\$1.12 / \$3.52 (FP8)	—	1M	Yes	Cheapest blended (~\$0.72)
Wafer	\$1.20 / \$4.10	—	1M	Yes	Fast cheap (>200 t/s); FP4
Fireworks AI	\$1.40 / \$4.40 (FP8)	\$0.55 / \$2.19	1M / 203K	Yes	Fastest (315 t/s); fixed 6,000 RPM ceilin,
Together AI	\$1.40 / \$4.40 (FP4); cached \$0.26	\$1.00 blended	1M (reseller page lists 256K)	Yes	LoRA fine-tune, downloadable weights
Novita AI	FP8 tier (~\$1.40 / \$4.40 class)	\$0.60 / \$2.20	200K-1M	Yes	<code>base_url=https://api.novita.a</code>
SiliconFlow	FP8 tier	\$0.50 / \$1.90 (page lists \$0.39/\$1.90)	205K	Yes (no JSON mode)	JSON mode missing — blocker for struc output
Parasail	FP8 tier	—	1M	Yes	

Provider	GLM-5.2 input/output (per 1M)	GLM-4.6 input/output	Context	OpenAI- compat	Notes
Baseten / CoreWeave / Nebius / FriendliAI / Databricks / Makora	~\$1.40 / \$4.40 class (varies; Nebius highest ~\$1.70 blended)	—	1M	Yes	FriendliAI supports both OpenAI + Ant

GLM-5.2 (max) blended price ranges ~\$0.72 (GMI) to \$1.70 (Nebius) per 1M on a 7:2:1 mix. FP4 routes (DeepInfra, Wafer) trade some quality for price; Z.ai/Fireworks/Novita serve FP8.

Artificial Analysis + 2

3. Free access options

- **OpenRouter:** No `:free` variant for GLM-5.2, GLM-5, or GLM-4.6. **GLM-4.5-Air is available free** at slug `z-ai/glm-4.5-air:free` (confirmed live on OpenRouter; 131K context, up to 96K output). Free-tier limits: 20 req/min; 50 req/day under \$10 lifetime credit, 1,000 req/day after a one-time \$10 purchase. `Developers Digest + 2`
- **Z.ai / BigModel free tiers:** New BigModel (`open.bigmodel.cn`) accounts get **20,000,000 free tokens** automatically — the console states "New users get 20.00 million tokens for free!" (plus ~120 image/video credits). **Zhipu's ZCode 3.0 provides 3 million free GLM-5.2 tokens per day for eligible users** (per Avenchat, June 22 2026). Z.ai's three always-free Flash models (GLM-4.7-Flash, GLM-4.5-Flash text; GLM-4.6V-Flash vision) are genuinely free but rate-limited and tuned for speed over depth.
- **Self-hosting (MIT):** Weights at `huggingface.co/zai-org/GLM-5.2` and ModelScope; FP8, BF16, NVFP4 (NVIDIA), W4AFP8 (Phala) and GGUF (Unsloth) builds exist. Zero per-token cost. `Hugging Face`
- **GLM Coding Plan launch promo:** First-month promo rates were as low as \$3 (Lite) / \$15 (Pro) at launch week (now expired/stepped up).
- **Other free/near-free routes (mostly coding-tool-bound):** OpenCode Go referral credits, Devin Pro bundling, and a limited Hugging Face Inference Providers window. There is **no permanent, unlimited free hosted GLM-5.2 API**.

4. GLM Coding Plan subscriptions

Flat-fee, prompt-quota subscription usable only inside ~20+ supported coding tools (Claude Code, Cline, Roo Code, Cursor, OpenClaw, Kilo Code, etc.). All tiers include GLM-5.2, GLM-5-

Turbo, GLM-4.7, GLM-4.5-Air.

Tier	Standard /mo	Promo / annual	Quota
Lite	\$18	~\$12.60/mo (yearly); \$3 launch promo	~80 prompts / 5h, ~400/week, 100 MCP calls/mo
Pro	\$72	~\$50.40/mo (yearly); \$15 launch promo	~400 prompts / 5h, ~2,000/week, 1,000 MCP calls/mo (5× Lite)
Max	\$160	~\$112/mo (yearly)	~1,600 prompts / 5h, ~8,000/week, 4,000 MCP calls/mo (20× Lite)

Quotas reset on 5-hour rolling windows. GLM-5.2 consumes ~3× quota at peak / 2× off-peak vs lighter models. The Coding Plan ships both OpenAI- and Anthropic-compatible endpoints, but the OpenAI URL must be the coding-only endpoint (`https://api.z.ai/api/coding/paas/v4`) (not the general `/api/paas/v4`). **Important:** the Coding Plan license forbids use through unauthorized third-party tools / custom integrations — for mindX's API backend you must use the standalone pay-per-token API, not the Coding Plan. (LushBinary)

5. Self-hosting requirements

- **GLM-5.2 (753B/40B MoE):** ~744–753 GB VRAM at FP8 (weights only), ~1.49 TB at BF16, ~372 GB at INT4 — plus KV cache for 1M context and ~10–20% overhead. Practically **8×H200** (~1,128 GB aggregate) at FP8, ~16 GPUs at BF16. NVFP4 build targets B200/B300; W4AFP8 needs Hopper (SM90). Datacenter-class; not a laptop/prosumer model. (Hugging Face) (Hugging Face)
- **GLM-4.5-Air (106B/12B MoE, 128K context):** runs on **2×H100** or **1×H200** at FP8. The practical self-host sweet spot for cost-sensitive teams; ~2× faster than full-size models.
- **GLM-4.6 (355B/32B, 200K):** ~8×H200 FP8 / 16×H100 territory.
- Serve with (`vllm serve zai-org/GLM-5.2 --tensor-parallel-size 8 --tool-call-parser glm47 --reasoning-parser glm45`) (or SGLang with EAGLE speculative decoding). vLLM prefers power-of-two tensor-parallel sizes; size for the memory budget, then round up. (Hugging Face)

Details

Integration into mindX (OpenAI-compatible router, JWT-gated, 350+ routes)

mindX already exposes an OpenAI-compatible surface, so GLM integration is a base-URL + key + model-slug change. Recommended wiring:

OpenAI-compatible (lead with this — matches mindX's existing stack):

```
base_url = "https://api.z.ai/api/openai/v1" # or https://api.novita.ai/openai,
                                              # https://openrouter.ai/api/v1,
                                              #
https://api.deepinfra.com/v1/openai
api_key = <provider key> # store server-side, never expose to
JWT clients
model = "glm-5.2" # Z.ai; "zai-org/GLM-5.2" on
Novita/Together/DeepInfra;
# "z-ai/glm-5.2" on
OpenRouter
```

For the 1M-context mode in some tools, the model id is `glm-5.2[1m]`.

Anthropic-compatible (only if a route needs Claude-style Messages API):

```
ANTHROPIC_BASE_URL = "https://api.z.ai/api/anthropic"
ANTHROPIC_AUTH_TOKEN = <z.ai key>
```

Z.ai is the only major provider besides Anthropic offering a true Anthropic-compatible endpoint; FriendliAI also supports both OpenAI and Anthropic Messages formats.

Multi-route, JWT-gated considerations:

- Keep provider API keys server-side in mindX; mint your own JWTs for downstream callers and never pass the upstream GLM key through. GLM endpoints authenticate with a simple bearer key, so a thin proxy route inside mindX is the clean pattern.
- Set a long client/server timeout: first-token latency for 1M-context GLM calls is reported at 30–90s. The Z.ai docs recommend `API_TIMEOUT_MS=3000000` for long calls — apply an equivalent at your gateway so JWT-gated routes don't kill long agentic loops.
- For BDI/agentic loops that resend a large stable system prompt or world-state, exploit cached input (\$0.26/1M on Z.ai, ~\$0.18–0.205/1M on DeepInfra) — it cuts repeated-context cost ~80%. `(TogetherAI + 2)`
- GLM-5.2 has thinking enabled by default (reasoning tokens billed at output rate). Disable per-turn with `reasoning={"enabled": False}` for cheap/simple routes; reserve High/Max effort for hard tasks. `(TogetherAI)`
- For resilience across 350+ routes, OpenRouter (auto-router + automatic failover across 22+ hosts) or a small internal fallback list (Z.ai → DeepInfra → Fireworks) avoids single-provider outages. Avoid SiliconFlow for structured-output routes — it lacks JSON mode. `(OpenRouter)`

Recommendations

Stage 1 — Prototype (this week, near-zero cost):

- Sign up at BigModel (20,000,000 free tokens) or use Z.ai's free Flash models (GLM-4.7-Flash / GLM-4.5-Flash) to validate the OpenAI-compatible wiring into mindX.
- For free experimentation through one key, add `z-ai/glm-4.5-air:free` via OpenRouter (20 req/min, 50-1,000 req/day). [\(Klymentiev\)](#)

Stage 2 — Low-volume production:

- Use Z.ai first-party OpenAI-compatible API with GLM-5.2 (\$1.40/\$4.40) for quality-critical routes and GLM-4.7 or GLM-4.5-Air (\$0.60/\$2.20 and \$0.20/\$1.10) for routine routes. Turn on prompt caching for your stable BDI system prompt.
- If you want passthrough pricing + failover, route through OpenRouter or DeepInfra (FP4, cheapest output) instead.

Stage 3 — Scale / cost optimization:

- Tier your model routing: GLM-4.5-Air or GLM-4.7-Flash for classification/cheap turns, GLM-5.2 only for hard reasoning/coding. This is the single biggest cost lever (a workload can be ~5× cheaper on Air than on 5.2).
- Benchmark FP4 (DeepInfra/Wafer, ~\$0.72-0.95 blended) vs FP8 (Z.ai/Fireworks) on your own tasks before optimizing purely on price.

Stage 4 — Self-host (only at high, steady volume or strict data residency):

- Self-host GLM-4.5-Air on 1×H200/2×H100 for the best cost/effort balance; reserve full GLM-5.2 self-hosting (8×H200) for genuinely high sustained throughput or air-gapped compliance.

Thresholds that change the recommendation:

- If monthly standalone-API token spend exceeds ~\$72-160 and your usage is coding-tool-shaped, the GLM Coding Plan is cheaper — but only if you can run inside a supported tool (not a custom backend).
- If sustained throughput exceeds what amortized 8×H200 rental covers vs per-token billing, self-hosting wins.
- If a route needs vision, switch to GLM-4.6V (\$0.30/\$0.90) or the free GLM-4.6V-Flash.

[\(Siliconflow\)](#)

Caveats

- **Pricing volatility:** GLM pricing and Coding Plan tiers change frequently (Z.ai notes "prices refreshed weekly"; launch promos have already expired). Verify at [z.ai/subscribe](#),

docs.z.ai/guides/overview/pricing, and openrouter.ai before committing.

- **Parameter-count conflict:** Official zai-org card says 753B total; an arXiv survey and several trackers cite 744B (the GLM-5 backbone figure). 40B active is consistent across sources.
- **Benchmark claims are largely vendor-reported.** Z.ai initially published no full benchmark suite for GLM-5.2; the SWE-bench Pro / Terminal-Bench numbers are self-reported or early third-party and await broad independent verification.
- **Coding Plan ≠ API backend.** The Coding Plan's terms restrict use to officially supported coding tools; using its quota through mindX's custom API would violate terms. Use the standalone pay-per-token API for backend integration.
- **Quantization affects quality.** The cheapest FP4 routes can differ subtly from FP8/BF16 on agentic/coding tasks — test on your workload. ([Developers Digest](#))
- **Latency & region:** Z.ai servers are primarily in China (~100–200ms from US/EU; slower during Chinese business hours). For latency-sensitive routes, a US-hosted aggregator (Fireworks, Together, DeepInfra) may be preferable.
- **Reseller-page errors exist:** e.g., Together AI's GLM-5.2 page lists "256K context / 131,072 output," contradicting the official 1M-context spec. Prefer first-party sources (Z.ai docs, Hugging Face, OpenRouter, provider docs) where they conflict with secondary trackers.